

Mohamed Baha Riahi

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PROFILE

Final-year ICT engineering student at Sup'Com (graduating 2027) building LLM infrastructure, RAG systems, and agentic applications. Seeking an **end-of-studies internship (PFE) in AI Engineering** starting early 2027.

EXPERIENCE

AI Engineering Intern

Jul 2026 – Aug 2026

Elyadata

Tunis, Tunisia

- Built a **semantic caching layer for LLM applications** that answers paraphrased queries from a vector cache (Redis HNSW, 1024-dim embeddings, cosine similarity), cutting response time from 23–75s to about 1.5s on cache hits: up to **41× faster**, with a hit rate up to **33%** on real LMSYS-Chat queries.
- Tuned the similarity threshold on **10,000 Quora duplicate-question pairs** (0.50–0.95 sweep) and chose 0.80 for the best F1 (68.3%; precision 63%, recall 74%), balancing wrong cached answers against missed reuse.
- Added a **PII guardrail** with Microsoft Presidio: queries containing personal data skip embedding and caching entirely, and custom entity recognizers hot-reload through an authenticated API.
- Routed models through LiteLLM to local Ollama (Qwen) to stay model-agnostic; compared 3 invalidation and 3 eviction policies, with 37 tests.

Keywords: LLM Infrastructure, Semantic Caching, Vector Search, Redis, pgvector, LiteLLM, Presidio

AI & Cybersecurity Engineering Intern

Jul 2025 – Aug 2025

SFM Technologies

Tunis, Tunisia

- Explored **AI-assisted security analytics**: honeypot dataset analysis, attack visualization, and LLM-based log analysis (Gemini API) to extract insights from security events.
- Designed and deployed a **segmented honeypot network** with an end-to-end monitoring pipeline (pfSense segmentation, traffic routing, centralized log collection) to capture brute-force and reconnaissance attacks.

Keywords: LLM Log Analysis, Gemini API, Security Analytics, Honeypots, Wazuh, pfSense

PROJECTS

Internal Coding Agent (SWE-Agent)

Sep 2026 – Present

- Building an **agentic coding system** on Open-SWE (LangGraph + Deep Agents) that explores repositories, implements fixes, runs tests, and opens pull requests for simple bug fixes and dependency updates.
- Integrating **tool use and sandboxed code execution**, with a self-hosted LiteLLM gateway (Docker Compose, PostgreSQL) for model routing.

Keywords: Agentic AI, LLM Agents, Tool Use, LangGraph, Deep Agents, LiteLLM, GitHub API, Sandboxing

Malware Detection with Deep Learning

Feb 2026 – May 2026

- Developed and compared **4 deep learning architectures** (CNN, LSTM, GNN, Transformer) for malware classification from dynamic API call sequences (300+ samples), with a pipeline for training, evaluation, and model comparison.

Keywords: Deep Learning, PyTorch, CNN, LSTM, GNN, Transformers, Sequence Classification

AthenaAI: RAG Study Companion

Nov 2025 – Jan 2026

- Built an end-to-end **RAG application** on a local LLM (Gemma 7B via Ollama): users upload PDFs, ask questions in a multi-turn conversation, and generate flashcards through **structured JSON output**.
- Built the **retrieval pipeline**: Marker PDF-to-Markdown conversion (layout-aware, OCR), Markdown-aware chunking, MiniLM sentence embeddings, and **ChromaDB semantic search** to ground answers in the most relevant passages.

Keywords: RAG, Semantic Search, Embeddings, ChromaDB, LangChain, Ollama, Structured Output

EDUCATION

Higher School of Communications of Tunis (Sup'Com)

Ariana, Tunisia

Engineering Degree in Information and Communication Technologies (ICT)

2024 – 2027 (expected)

- Relevant coursework:** Data Structures & Algorithms, Distributed Systems, Microservices Architecture, UML & Software Design

Preparatory Institute for Engineering Studies of Tunis (IPEIT)

Tunis, Tunisia

Preparatory Studies in Mathematics and Physics

2022 – 2024

- Relevant coursework:** Linear Algebra, Probability, Calculus, Scientific Simulation, Numerical Methods, Relational Databases

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Bash

LLM & AI: LLMs, RAG, Agentic AI, Embeddings, Vector Search, Semantic Caching, NLP, Deep Learning

Frameworks: PyTorch, LangGraph, LangChain, Sentence Transformers, FastAPI

AI Infrastructure: LiteLLM, Ollama, Redis Stack, pgvector, PostgreSQL, ChromaDB, Microsoft Presidio

DevOps & Tools: Docker, Docker Compose, Git, Linux, Prometheus, Grafana, pytest